WORKED SOLUTION

STEP II 1990, Question 7

Pure • Differential Equations

STEP II 1990, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q7

The key idea

An exponential trial solution turns a delay equation into an ordinary equation for p, because a time shift just multiplies e^{pt} by a constant. The equation $(*)$ mixes p with e^{-p}, so it can't be solved exactly, but multiplying through by e^{p} puts all the p-dependence on one side: the roots are where the single curve $y = (p+1)e^{p}$ meets the horizontal line $y = k$. Counting roots then comes down to sketching one curve.

Solution

The equation for p

Put $f(t) = Ae^{pt}$ with $A \neq 0$ (we want non-zero solutions). Then $f'(t) = pAe^{pt}$ and $f(t-1) = Ae^{p(t-1)} = e^{-p}\,Ae^{pt}$, so $(\dagger)$ becomes

$$Ae^{pt}(p+1-ke^{-p}) = 0 \quad \text{for all } t.$$

Since $A \neq 0$ and $e^{pt} > 0$, this holds for all t if and only if

$$p+1 = ke^{-p},$$

which is $(*)$. So such solutions exist exactly when p is a real root of $(*)$, and then any A works.

How many real roots?

Multiplying $(*)$ by $e^{p} > 0$ gives the equivalent equation

$$g(p) = k, \qquad \text{where } g(p) = (p+1)e^{p}.$$

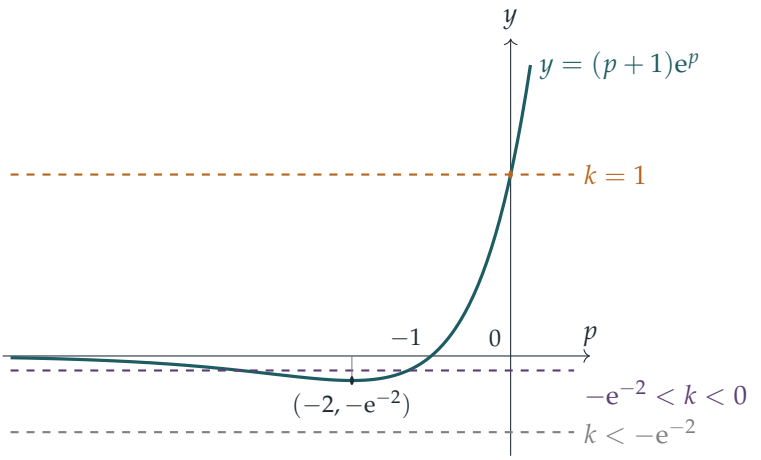
So we count the intersections of the curve $y = g(p)$ with the horizontal line $y = k$.

Ask yourself What does $y = (p+1)e^p$ look like? Where is it smallest, and what happens at each end?

Differentiating, $g'(p) = e^p + (p+1)e^p = (p+2)e^p$, which is negative for $p < -2$ and positive for $p > -2$. So g decreases on $(-\infty, -2]$, increases on $[-2, \infty)$, and has its global minimum

$$g(-2) = -e^{-2}.$$

At the ends: $g(p) \to \infty$ as $p \to \infty$, and $g(p) \to 0$ as $p \to -\infty$ (exponential decay beats linear growth); for $p < -1$ the factor $p+1$ is negative, so g approaches 0 from below. The curve crosses the axis only at $p = -1$, and $g(0) = 1$.



Reading off the picture (and the monotonicity above, which makes it rigorous):

- $k > 0$: on $(-\infty, -1]$ we have $g \le 0 < k$, and on $(-1, \infty)$ g increases strictly from 0 to ∞, so there is **exactly one** root, and it satisfies $p > -1$.
- $-e^{-2} < k < 0$: on $(-\infty, -2]$, g decreases strictly from values near 0 down to $-e^{-2}$, so it takes the value k once; on $[-2, -1]$ it increases strictly from $-e^{-2}$ to 0, so it takes the value k once more; for $p > -1$, $g > 0 > k$. Hence **two** roots, one in $(-\infty, -2)$ and one in $(-2, -1)$.
- $k = -e^{-2}$: the line touches the curve at its minimum, giving the **single** root $p = -2$.
- $k < -e^{-2}$: the line lies below the whole curve, so there are **no** roots.

So $(*)$ has 0, 1 or 2 real roots depending on k, as required, and solutions of the form Ae^{pt} exist precisely when

$$k \ge -e^{-2}, \quad k \ne 0.$$

(The value $k = 0$ is excluded by the question; it would give the single root $p = -1$.)

When do the solutions decay?

Ask yourself Which p give $Ae^{pt} \to 0$, and how do we locate the roots relative to 0 without solving $(*)$?

With $A \neq 0$, $Ae^{pt} \to 0$ as $t \to +\infty$ if and only if $p < 0$. We never need the roots explicitly, only their position relative to 0.

- If $-e^{-2} \leq k < 0$, every root lies in $(-\infty, -1)$ (indeed $g \geq 0$ for $p \geq -1$, so $g(p) = k < 0$ forces $p < -1$). All such solutions decay.
- If $k > 0$, the unique root lies where g is strictly increasing, so $p < 0 \iff g(p) < g(0) \iff k < 1$. For $k = 1$ the root is $p = 0$, giving the constant solution $f = A$, which does not tend to zero; for $k > 1$ the root is positive and the solution grows.

Therefore the solutions tend to zero as $t \to +\infty$ exactly when

$$-e^{-2} \leq k < 1, \quad k \neq 0.$$

Checks

- $k = 1$: $p = 0$ satisfies $0 + 1 = 1 \cdot e^0$, and $f = A$ clearly solves $f' + f - f(t-1) = 0$. This is the borderline between decay and growth, as it should be.
- $k = -e^{-2} \approx -0.1353$: $p = -2$ gives $-1 = -e^{-2}e^2$. Numerically, $k = -0.13$ gives two roots, $p \approx -2.311$ and $p \approx -1.742$, either side of -2, while $k = -0.14$ gives none.
- Physical sense: with no feedback ($k \to 0$) the system is $f' = -f$, with decay rate $p = -1$, which is where the root for small $|k|$ sits. Strong positive feedback ($k > 1$) overcomes the damping and the solution grows.

Common slips

- Sketching $y = p + 1$ against $y = ke^{-p}$ and missing that for $k < 0$ the line can cut the curve twice or not at all; the tangency condition is easier to find with $g(p) = (p+1)e^p$.
- Forgetting the endpoint $k = -e^{-2}$, which does give a (repeated) root $p = -2$ and hence a decaying solution.
- Including $k = 1$ in the decay set: $p = 0$ gives a constant, not a decaying, solution.

Beyond STEP

Equation $(\dagger)$ is a *delay differential equation*, and $(*)$ is its characteristic equation. Unlike an ordinary linear ODE, it has infinitely many *complex* roots p, and the general solution

is built from all of them, so full stability needs every complex root to have negative real part; this is studied in control theory. The real roots of $(p+1)\mathrm{e}^p = k$ are values of the Lambert W function: $p+1 = W(k\mathrm{e})$, and $-\mathrm{e}^{-1}$ (here $k\mathrm{e} = -\mathrm{e}^{-1}$) is exactly the branch point of W. For linear ODEs and characteristic equations, see Boyce and DiPrima, *Elementary Differential Equations*.

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